HealthTrack: Predicting Hospital Readmission

Mini Project Report – Data Science

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1. Introduction

Hospital readmission prediction is a critical problem in healthcare analytics. HealthTrack is a data-driven system that uses patient clinical data collected at discharge to estimate the probability of readmission within 30 days. The system is built using Python and Streamlit and is designed for use as an academic demonstration of applied machine learning in healthcare.

2. Dataset Description

The dataset consists of 2,000 synthetically generated patient records with 10 features. The data was generated using controlled probability distributions to simulate realistic clinical patterns.

Feature	Type	Description
age	Numerical	Patient age in years
gender	Categorical	Male / Female / Other
num_prior_visits	Numerical	Hospital visits in last 1 year
length_of_stay	Numerical	Duration of current stay (days)
diagnosis	Categorical	Primary diagnosis at admission
discharge_type	Categorical	How the patient was discharged
insurance	Categorical	Insurance type
num_medications	Numerical	Medications at discharge
num_comorbidities	Numerical	Number of other conditions
readmitted_30days	Binary	Target: 1=Readmitted, 0=Not

3. Methodology

The project follows a standard supervised machine learning pipeline:

1. Data Generation – Synthetic patient records created using NumPy.
2. Exploratory Data Analysis – Distributions, correlations, and class balance examined.
3. Preprocessing – Categorical variables label-encoded; no imputation required (no missing values).
4. Model Selection – Random Forest Classifier chosen for its robustness and interpretability.
5. Training – 80/20 train-test split with stratification; class_weight='balanced' used.
6. Evaluation – Accuracy, ROC-AUC, precision, recall, F1-score, 5-fold cross-validation.
7. Deployment – Streamlit web app with prediction form, insight charts, and model report.

4. Model Configuration

Parameter	Value
Algorithm	Random Forest Classifier
n_estimators	150
max_depth	8

min_samples_leaf	10
class_weight	balanced
random_state	42

5. Results

Metric	Value
Test Accuracy	82.0%
ROC-AUC Score	0.899
5-Fold CV Acc.	82.8%
Precision (Readmitted)	0.83
Recall (Readmitted)	0.81
F1-Score (Readmitted)	0.82

The most important features identified by the model were: number of prior hospital visits (32.3%), patient age (25.0%), and length of stay (15.0%). Discharge type and insurance had lower but non-trivial importance scores.

6. Key Findings

- Prior hospital visits is the single strongest predictor of readmission.
- Elderly patients (age 65+) have significantly higher readmission rates.
- Patients with Heart Failure, Sepsis, or COPD have the highest diagnosis-specific readmission rates.
- Patients discharged 'Against Medical Advice' have an 83% readmission rate in the dataset.
- The number of comorbid conditions positively correlates with readmission risk.

7. Conclusion

HealthTrack successfully demonstrates the application of Random Forest classification to hospital readmission prediction. The system achieves strong predictive performance (82% accuracy, 0.899 AUC) and provides an intuitive interface for clinical use. Future improvements could include training on real EHR data, integrating SHAP values for explainability, and adding time-series features from patient history.

8. References

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